Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the f1 score by dataset.

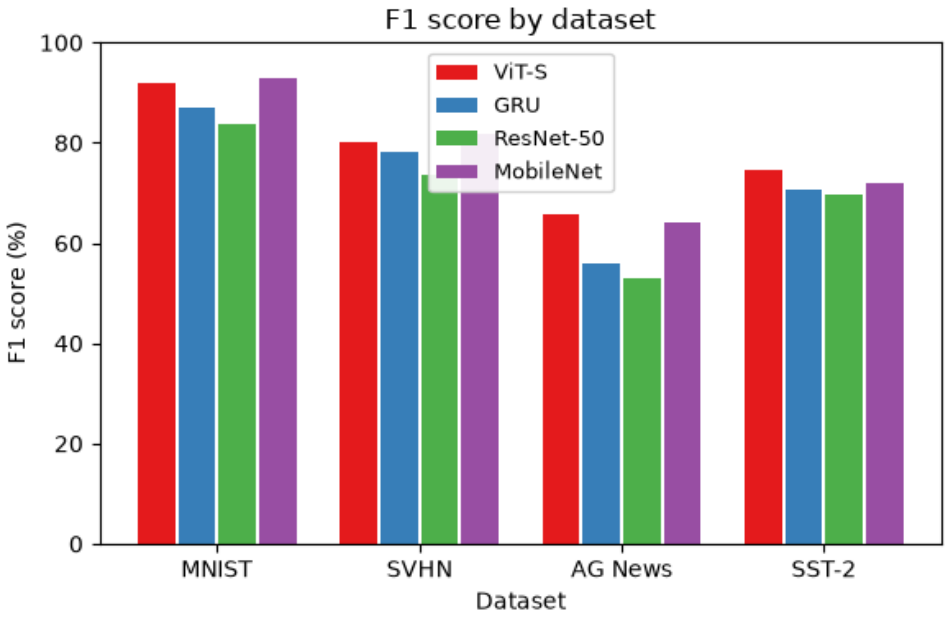


Figure 1: F1 score by dataset for the models in this experiment.

As shown in Figure 1, on SST-2, MobileNet is 4.4 points better than ResNet-50. ViT-S outperforms GRU by 1.9 percentage points on SVHN. ViT-S reaches 80.0% F1 score on SVHN. On AG News, GRU is 3.1 points better than ResNet-50.

Figure 2 shows the accuracy during training.

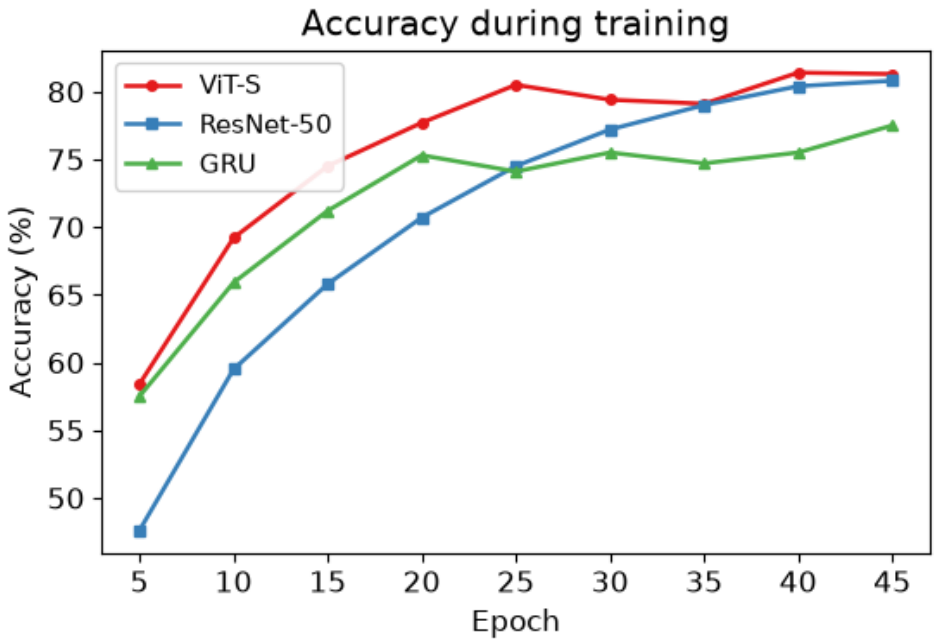


Figure 2: Accuracy during training for the models in this experiment.

As shown in Figure 2, GRU reaches 76.0% accuracy at epoch 20. ResNet-50 reaches 79.1% accuracy at epoch 35.

Figure 3 shows the f1 score during training.

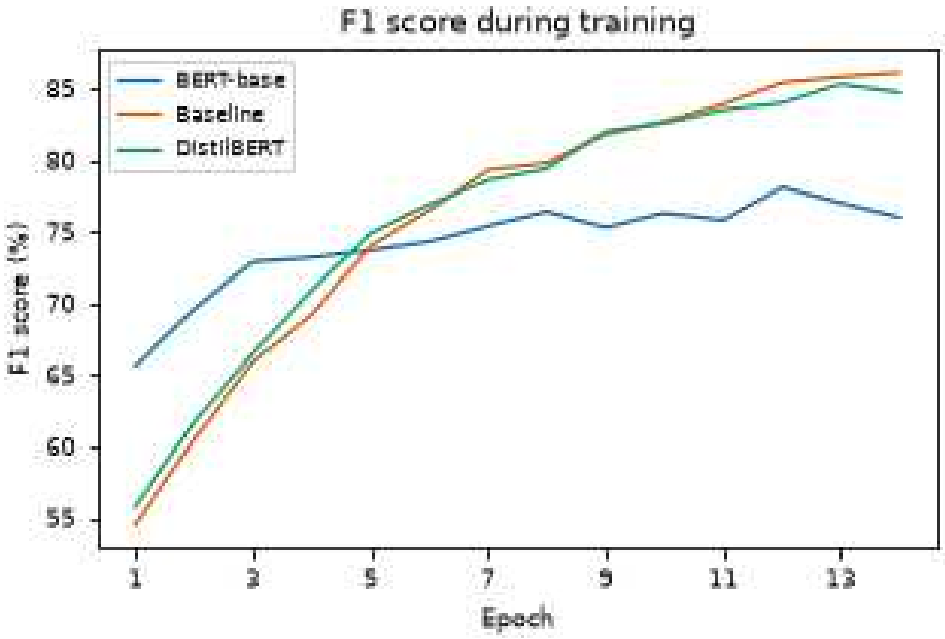


Figure 3: F1 score during training for the models in this experiment.

As shown in Figure 3, from epoch 6 to epoch 12, the F1 score of DistilBERT improves by a relative 9%. At epoch 4, BERT-base is 4.0 points better than Baseline (Figure 3). Baseline attains a F1 score of 79.8% at epoch 8.

3 Discussion

The differences between the models are discussed in the next section.